

1 Data and partitioning

ISIC 2018 Task 3 training set

10,015 dermoscopic images
7,470 distinct lesions, 7 classes

Lesion-level partitioning

StratifiedGroupKFold ($k = 5$)
group = lesion_id,
stratified by diagnosis;
no lesion in more than one fold

SC-EfficientNet pilot (fold 0, B0)

pre-specified pass criteria

2 Models and training

Two scales

B0: 224×224 , CE, no sampler
B4: 384×384 , CE, weighted sampler

Five model variants

- Standard baseline (1×1 head conv)
- Headless control
- Attention-only (B0 only)
- KL-only ablation (mask via KL only)
- SC-EfficientNet (supervised mask + priors)

Training protocol

Adam, lr 10^{-4} , 20 epochs, FP32
seeds 42, 43, 44; final-epoch model

3 Out-of-fold evaluation

Pooled OOF predictions

per configuration and seed
10,015 held-out images

Metrics and paired tests

accuracy, balanced accuracy,
macro-F1 (mean $\pm$ SD over seeds);
lesion-level bootstrap 95% CI;
exact McNemar test

Mask diagnostics

raw mask mean and fraction
of positions < 0.5

4 Additional B4 analyses

Official ISIC 2018 test set

1,512 images; 3 seeds

Synthetic artifacts

hair, ruler, occlusion; 3 seeds

TTA and XAI

5 views; attention vs. Grad-CAM

ROC / PR, class-wise

SC-EfficientNet-B4, seed 42

Complexity and latency

params, FLOPs, GPU time